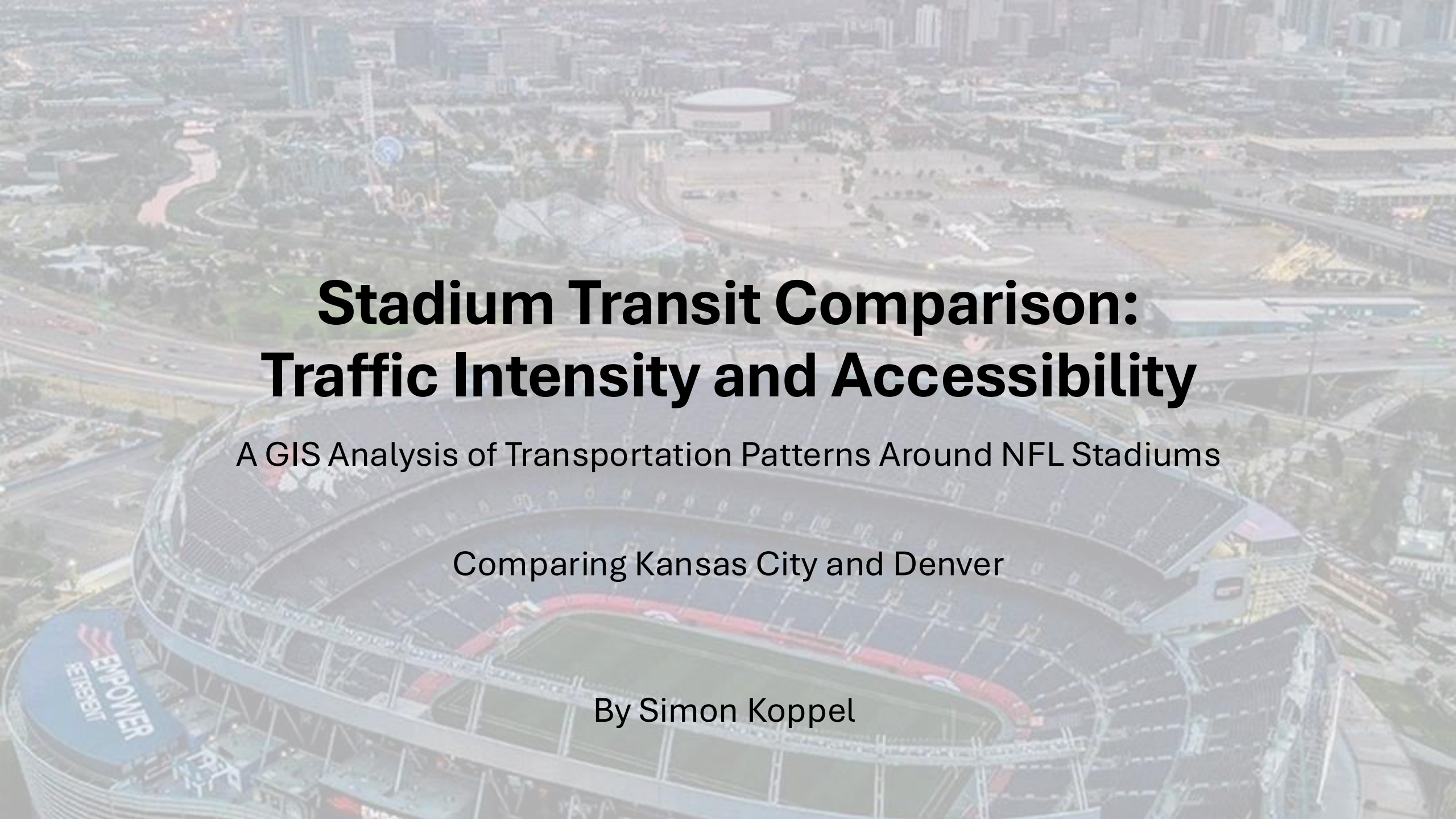


Stadium Transit Comparison: Traffic Intensity and Accessibility

A GIS Analysis of Transportation Patterns Around NFL Stadiums

Comparing Kansas City and Denver

By Simon Koppel

Project Overview

Purpose

Research Question

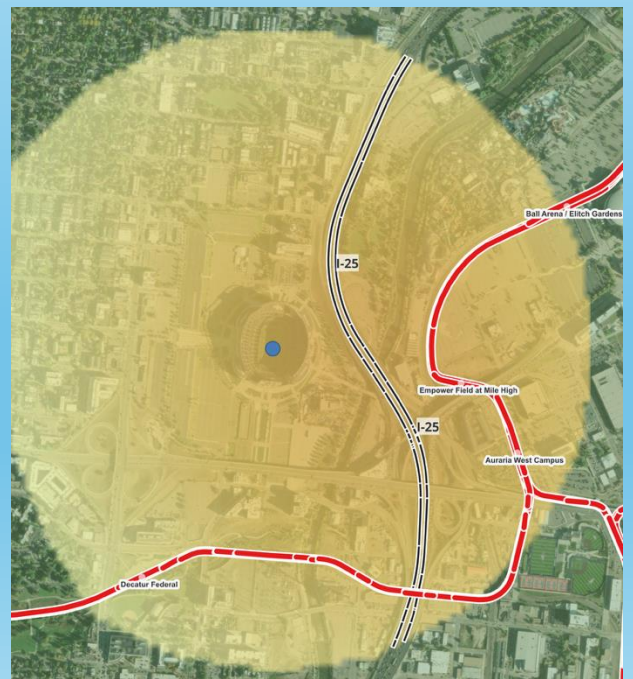
How does access to rapid transit influence traffic intensity and travel behavior around NFL stadiums?

- Evaluate how transit access impacts traffic near NFL Stadiums
- Compare transit-rich vs transit-limited environments
- Support planning for more sustainable stadium access

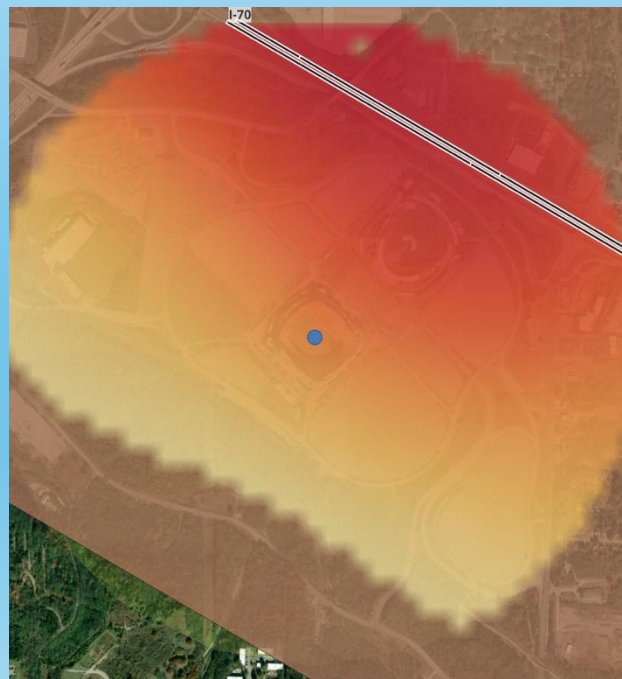
Study Areas

- Denver
- Kansas City

Denver (Transit Access)



Kansas City (Limited Transit)



Data and Methodology

Data Sources

Traffic Data (Average Annual Daily Traffic)

- Missouri Department of Transportation (MODOT)
- Denver Regional Council of Governments (DRCOG)

Transit Data

- OpenStreetMap (via QuickOSM in QGIS)
- Denver Light Rail Network

Base Layers

- Satellite imagery (ESRI / QGIS basemap)

Images

- *Empower Field at Mile High*. Visit Denver. https://visitdenver.com/listing/empower-field-at-mile-high/4559/
- *Denver Broncos Logo*. NFL. https://static.www.nfl.com/league/api/clubs/logos/DEN.svg
- *RTD 2025 Light Rail Ridership*. Regional Transportation District. https://www.rtd-denver.com/public-information-dashboards/ridership
- *RTD Light Rail Vehicle*. Greater Denver Transit. https://www.greaterdenvertransit.com/rtdlightrailvehicles/

Methodology

Data Preparation

- Imported traffic and road network data into QGIS
- Cleaned and projected datasets (WGS84/projected CRS)

Spatial Analysis

- Created buffers (5-8 mile radius) around each stadium.
- Extracted interstate and major road segments within buffers.

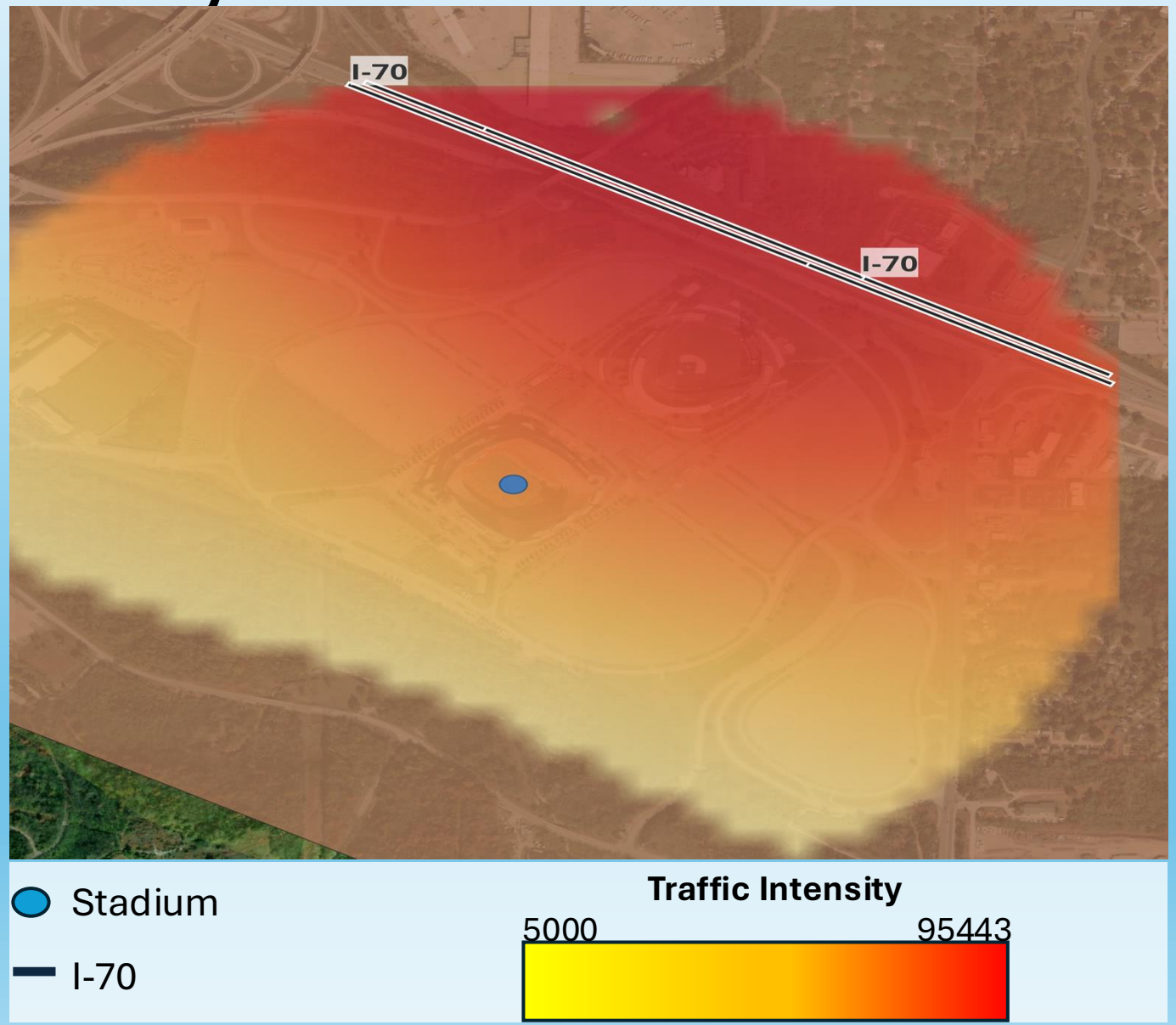
Traffic Visualization

- Applied graduated symbology to show traffic intensity.
- Compared congestion patterns near each stadium

Transit Comparison

- Mapped transit access (Denver vs Kansas City)
- Evaluated relationship between transit ability and traffic

Kansas City – Arrowhead Stadium Traffic Analysis



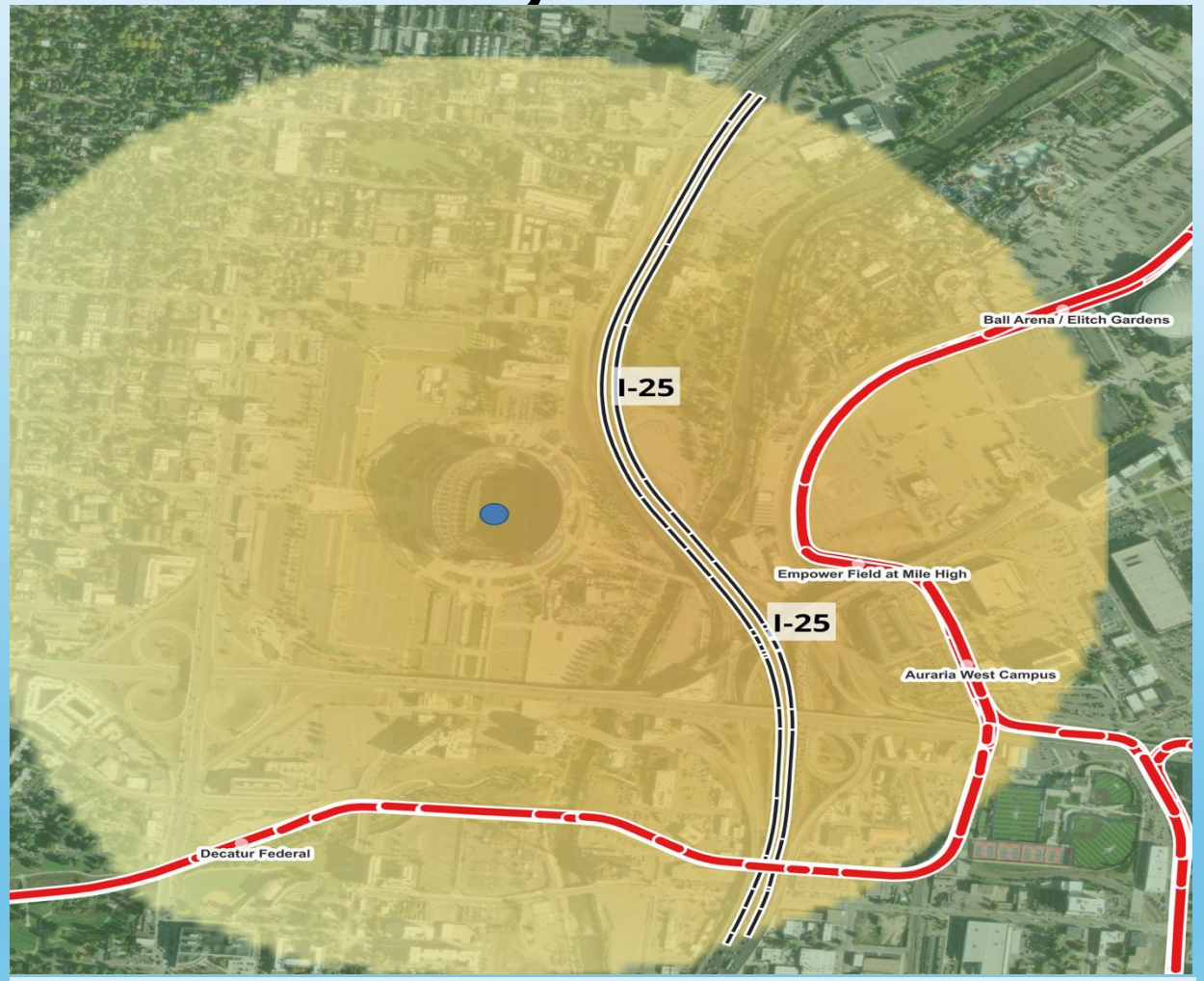
Traffic Patterns

- High traffic intensity concentrated along **Interstate 70**
- Congestion increases closer to the stadium
- Traffic radiates outward from stadium area

Transit Access

- No rapid transit serving the stadium
- Area is **highly car-dependent**
- Limited Alternatives to Driving

Denver-Empower Field at Mile High Stadium Traffic Analysis



Stadium
I-70
Light Rail

Traffic Intensity

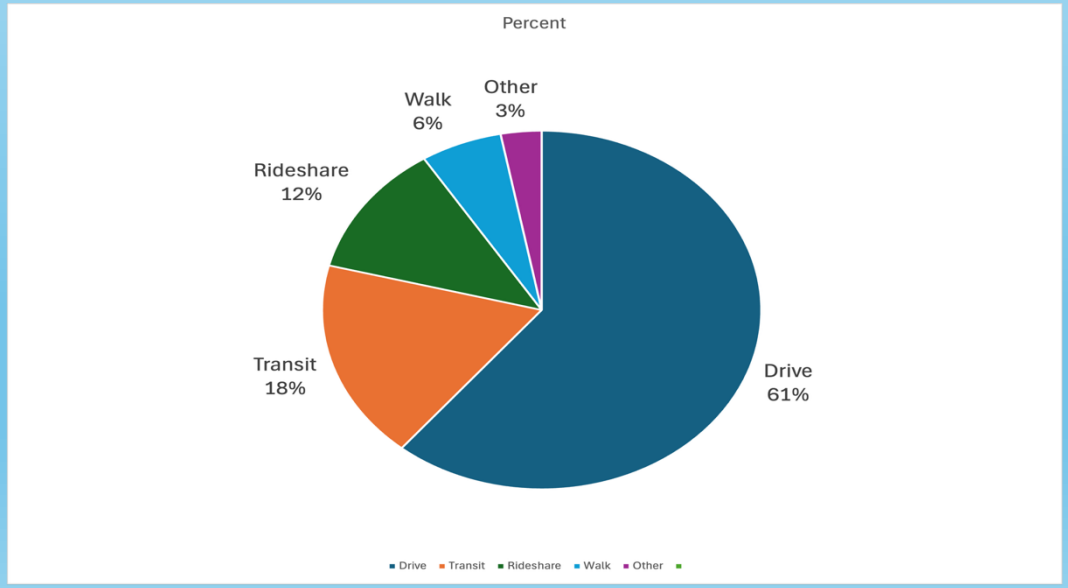
5000 95443

Traffic Patterns

- Traffic intensity is more **evenly distributed**
- Less concentrated congestion near the stadium
- Multiple access routes reduce bottlenecks

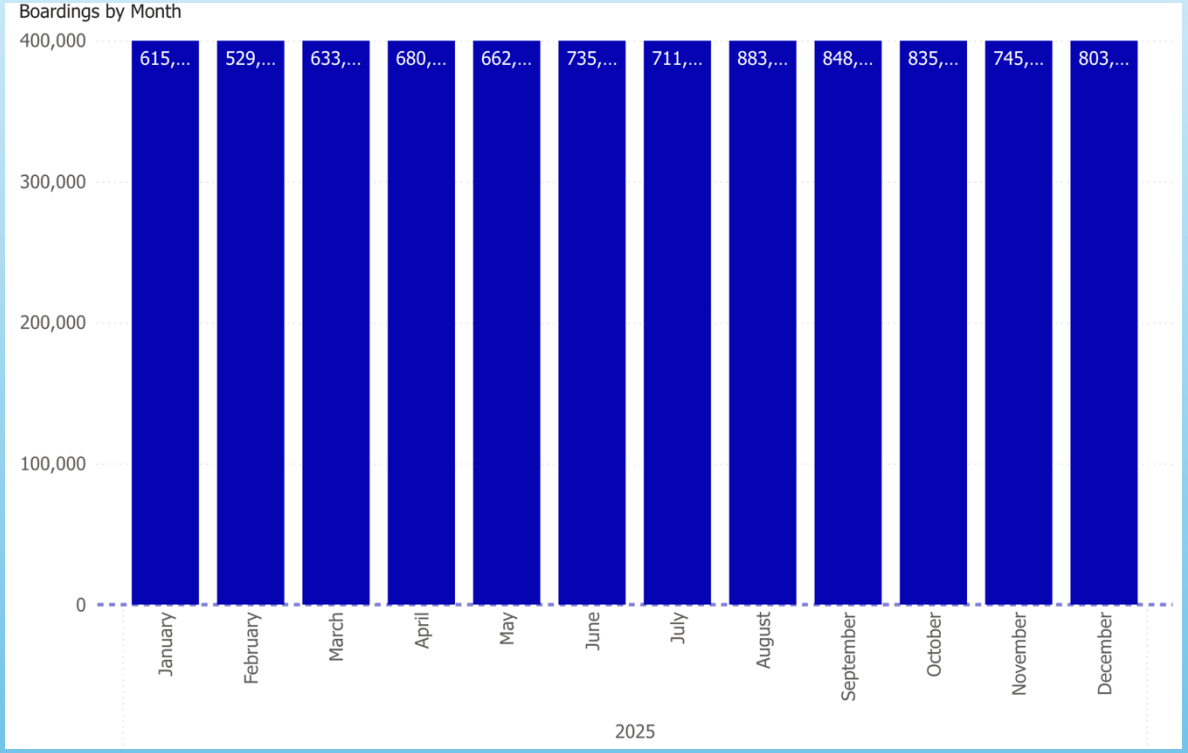
Transit Access

- Light rail provides **direct access to the stadium**
- Nearby stations improve accessibility
- Reduced dependence on driving



Fan Travel Mode-Denver (18 percent use transit!)

RTD Light Rail boardings by month (2025)



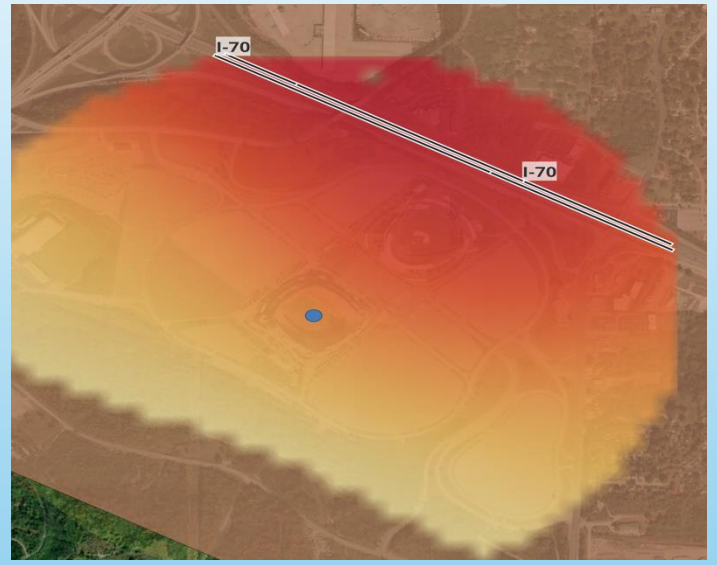
RTD Monthly Boardings (2025)

- Ridership increases during NFL season months (September-December)

RTD Light Rail Vehicle

Transit ridership peaks during football season, indicating that fans actively use light rail to access the stadium!

Kansas City



- **Car-dependent**
- Congestion concentrated near stadium
- Limited access routes

Denver

- **Transit-accessible**
- Traffic more distributed
- Multiple access options

Transit access fundamentally changes traffic patterns and travel behavior.

The clear winner in this transit themed 🏆 bout is.....

The Denver Broncos!!